Functional Analysis

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1. Metric spaces

Definition 1.1 (Metric space)**.** Let X be a set. A function $d\colon X \times X \to \mathbb{R}$ is a *metric* if for all $x, y, z \in X$ it satisfies $d(x,y) \geq 0$ with equality exactly when $x = y$, the symmetry $d(x,y) = d(y,x)$, and the triangle inequality $d(x,z) \leq d(x,y) + d(y,z)$. The pair (X,d) is a *metric space*.

Definition 1.2 (Cauchy sequence)**.** A sequence (x_n) in a metric space (X,d) is *Cauchy* if for every $\varepsilon > 0$ there is an $N \in \mathbb{N}$ such that $d(x_n, x_m) < \varepsilon$ for all $n, m > N$.

Definition 1.3 (Completeness)**.** A metric space (X,d) is *complete* if every Cauchy sequence in X converges to a limit lying in X.

Example 1.4 (Shortest-path metric)**.** Let $G = (V,E)$ be a connected graph and let $d(v_1, v_2)$ be the length of a shortest path between v_1 and v_2. Then d is a metric on V. Connectedness is what makes d finite.

Proposition 1.5 (Completion)**.** For every metric space (X,d) there is a complete metric space $(\overline{X}, \overline{d})$ together with a map $i\colon X \to \overline{X}$ such that $\overline{d}(i(x), i(y)) = d(x,y)$ for all $x, y \in X$ and $i(X)$ is dense in $\overline{X}$. It is unique up to isometric isomorphism: any two such spaces are related by a bijective isometry carrying one copy of X onto the other. $(\overline{X}, \overline{d})$ is the *completion* of (X,d).

Example 1.6. The space $(\mathbb{Q}, |\cdot|)$ is not complete; its completion is $(\mathbb{R}, |\cdot|)$.

Remark 1.7. It is common to suppress i and assume outright that $X \subset \overline{X}$, in which case the last condition reads: X is dense in $\overline{X}$.

2. Normed spaces and the p-norms

Definition 2.1 (Norm)**.** Let X be a vector space over $\mathbb{K}$, where $\mathbb{K}$ is $\mathbb{R}$ or $\mathbb{C}$. A function $\|\cdot\|\colon X \to \mathbb{R}$ is a *norm* if for all $x, y \in X$ and $\lambda \in \mathbb{K}$ it satisfies $\|x\| \geq 0$ with equality exactly when $x = 0$, the homogeneity $\|\lambda x\| = |\lambda| \|x\|$, and the triangle inequality $\|x + y\| \leq \|x\| + \|y\|$. Then $(X, \|\cdot\|)$ is a *normed space*, and $d(x,y) = \|x - y\|$ is a metric on X.

Example 2.2.

1. $X = \mathbb{K}^n$ with the p-norm $\|x\|_p = (\sum_{j=1}^{n} |x_j|^p)^{1/p}$ for $1 \leq p < \infty$, or with $\|x\|_\infty = \max_{1 \leq j \leq n} |x_j|$.
2. $X = C[0,1]$ with the supremum norm $\|f\|_\infty = \max_{t \in [0,1]} |f(t)|$, or with the p-norm $\|f\|_p = (\int_0^1 |f|^p)^{1/p}$.
3. $X = C^1[0,1]$ with $\|f\| = \max_t |f(t)| + \max_t |f'(t)|$.
4. $\ell_p = \left\{ x = (x_n) \;\middle|\; \sum_{n=1}^{\infty} |x_n|^p < \infty \right\}$ with $\|x\|_p = (\sum_{n=1}^{\infty} |x_n|^p)^{1/p}$.
5. ℓ_∞, the bounded sequences, with $\|x\|_\infty = \sup_n |x_n|$.

Of the norm axioms, only the triangle inequality for the p-norm is not immediate. The next two theorems supply it.

Theorem 2.3 (Hölder's inequality)**.** Let $1 < p < \infty$ and let q be its *conjugate exponent*, defined by $\frac{1}{p} + \frac{1}{q} = 1$. Then for all $x, y \in \mathbb{K}^n$,

$$\|x \cdot y\|_1 \;=\; \sum_{j=1}^{n} |x_j y_j| \;\le\; \Big(\sum_{j=1}^{n} |x_j|^p\Big)^{1/p} \Big(\sum_{j=1}^{n} |y_j|^q\Big)^{1/q} \;=\; \|x\|_p \|y\|_q, \tag{2.1}$$

where $x \cdot y = (x_1 y_1, \dots, x_n y_n)$. The inequality also holds for the endpoint pair $(p, q) = (1, \infty)$ and, by symmetry, for $(\infty, 1)$.

Remark 2.4. For $p = q = 2$ the inequality is the Cauchy–Schwarz inequality.

Proof. The endpoint case is immediate: $\sum_j |x_j y_j| \le (\max_j |y_j|) \sum_j |x_j| = \|x\|_1 \|y\|_\infty$.

Let now $1 < p < \infty$. The first step is Young's inequality: for all $a, b \ge 0$,

$$ab \le \frac{a^p}{p} + \frac{b^q}{q}. \tag{2.2}$$

If $a = 0$ or $b = 0$ the left side vanishes and the right side is non-negative, so assume $a, b > 0$ and put $u_1 = a^p > 0$, $u_2 = b^q > 0$, $\alpha_1 = \frac{1}{p}$, $\alpha_2 = \frac{1}{q}$, so that $\alpha_1 + \alpha_2 = 1$. The logarithm is concave on $(0, \infty)$, since $(\log)''(t) = -1/t^2 < 0$, and concavity applied to the convex combination $\alpha_1 u_1 + \alpha_2 u_2$ gives

$$\log\big(u_1^{\alpha_1} u_2^{\alpha_2}\big) = \alpha_1 \log u_1 + \alpha_2 \log u_2 \le \log(\alpha_1 u_1 + \alpha_2 u_2).$$

The logarithm is increasing, so $u_1^{\alpha_1} u_2^{\alpha_2} \le \alpha_1 u_1 + \alpha_2 u_2$, which is exactly (2.2), because $u_1^{\alpha_1} u_2^{\alpha_2} = (a^p)^{1/p} (b^q)^{1/q} = ab$.

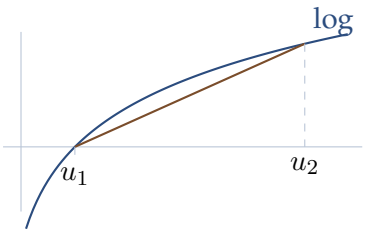


Figure 2.1. *Concavity of the logarithm: the chord over u_1 and u_2 lies below the graph, so $\alpha_1 \log u_1 + \alpha_2 \log u_2 \le \log(\alpha_1 u_1 + \alpha_2 u_2)$.*

The second step is a reduction by scaling. If $x = 0$ or $y = 0$ both sides of (2.1) vanish, so assume $\|x\|_p > 0$ and $\|y\|_q > 0$ and put $\tilde{x} = x/\|x\|_p$ and $\tilde{y} = y/\|y\|_q$, so that $\|\tilde{x}\|_p = \|\tilde{y}\|_q = 1$. Both sides of (2.1) are positively homogeneous of degree one in x and in y separately, so (2.1) for the pair x, y is equivalent to (2.1) for the normalised pair $\tilde{x}, \tilde{y}$. It therefore suffices to treat the case $\sum_j |x_j|^p = \sum_j |y_j|^q = 1$.

In that case, applying (2.2) with $a = |x_j|$ and $b = |y_j|$ and summing over j,

$$\sum_{j=1}^{n} |x_j y_j| \le \sum_{j=1}^{n} \Big(\frac{|x_j|^p}{p} + \frac{|y_j|^q}{q} \Big) = \frac{1}{p} + \frac{1}{q} = 1 = \|x\|_p \|y\|_q. \qquad \blacksquare$$

Theorem 2.5 (Minkowski's inequality)**.** Let $1 \le p < \infty$. Then $\|x + y\|_p \le \|x\|_p + \|y\|_p$ for all $x, y \in \mathbb{K}^n$.

Proof. For $p = 1$ the claim is the triangle inequality in $\mathbb{K}$ applied termwise. Let $1 < p < \infty$ and let q be the conjugate exponent, so that $q = \frac{p}{p-1}$ and hence $(p-1)q = p$. If $\sum_j |x_j + y_j|^p = 0$ the left side vanishes and the claim is trivial, so assume the sum is positive; being a finite sum, it is also finite. Using $|x_j + y_j| \le |x_j| + |y_j|$ and then Hölder (2.1) on each of the two resulting sums, with exponents q and p,

$$\begin{aligned}\sum_j |x_j + y_j|^p &\le \sum_j |x_j + y_j|^{p-1}|x_j| + \sum_j |x_j + y_j|^{p-1}|y_j| \\ &\le \Big(\sum_j |x_j + y_j|^{(p-1)q}\Big)^{1/q} \Big[\Big(\sum_j |x_j|^p\Big)^{1/p} + \Big(\sum_j |y_j|^p\Big)^{1/p}\Big] \\ &= \Big(\sum_j |x_j + y_j|^p\Big)^{1/q} \big(\|x\|_p + \|y\|_p\big).\end{aligned}$$

Dividing by the positive finite number $(\sum_j |x_j + y_j|^p)^{1/q}$ and using $1 - \frac{1}{q} = \frac{1}{p}$,

$$\|x + y\|_p = \Big(\sum_j |x_j + y_j|^p\Big)^{1/p} \le \|x\|_p + \|y\|_p. \qquad \blacksquare$$

2.1. Infinite sequences and functions

Both inequalities carry over from $\mathbb{K}^n$ to ℓ_p and to $C[0,1]$.

Theorem 2.6. Let $1 < p < \infty$ and $\frac{1}{p} + \frac{1}{q} = 1$. For all $x \in \ell_p$ and $y \in \ell_q$, $\|x \cdot y\|_1 \le \|x\|_p \|y\|_q$, and for all $f, g \in C[0,1]$,

$$\|fg\|_1 = \int_0^1 |fg| \le \Big(\int_0^1 |f|^p\Big)^{1/p} \Big(\int_0^1 |g|^q\Big)^{1/q} = \|f\|_p \|g\|_q.$$

Theorem 2.7. Let $1 \le p < \infty$. For all $x, y \in \ell_p$, $\|x+y\|_p \le \|x\|_p + \|y\|_p$, and for all $f, g \in C[0,1]$, $\|f + g\|_p \le \|f\|_p + \|g\|_p$.

2.2. The spaces $L_p[0,1]$

Definition 2.8 ($L_p[0,1]$)**.** Let $1 \le p < \infty$. On $C[0,1]$ the p-norm is a norm, but the space $(C[0,1], \|\cdot\|_p)$ is not complete. Its completion, in the sense of Proposition 1.5, is denoted $L_p[0,1]$.

Remark 2.9. The space $L_p[0,1]$ has a concrete description. Its elements are represented by those measurable functions $f\colon [0,1] \to \mathbb{K}$ for which $\int_0^1 |f|^p < \infty$, and two such functions represent the same element exactly when they agree outside a set of measure zero.

Remark 2.10. More generally, let $(X, \mathcal{A}, \mu)$ be a measure space and $1 \le p < \infty$. Then

$$L_p(X, \mathcal{A}, \mu) = \Big\{ f\colon X \to \mathbb{K} \,\Big|\, f \text{ measurable}, \int_X |f|^p \mathrm{d}\mu < \infty \Big\}, \qquad \|f\|_p = \Big(\int_X |f|^p \mathrm{d}\mu\Big)^{1/p},$$

and $L_p(X, \mathcal{A}, \mu)$ is a Banach space.

Exercise 2.11. Let $1 \le p < p' < \infty$. Show that $\ell_p \subset \ell_{p'}$.

Solution. Let $x \in \ell_p$. The series $\sum_j |x_j|^p$ converges, so $|x_j| \to 0$, and there is an N with $|x_j| < 1$ for all $j > N$. For such j the larger exponent gives the smaller value, $|x_j|^{p'} \le |x_j|^p$, whence

$$\sum_{j>N} |x_j|^{p'} \le \sum_{j>N} |x_j|^p < \infty.$$

The remaining finitely many terms are finite, so $\sum_j |x_j|^{p'} < \infty$, that is $x \in \ell_{p'}$. In particular $\ell_1 \subset \ell_{3/2} \subset \ell_2 \subset \cdots \subset \ell_\infty$. ◊

Exercise 2.12. Let $1 \le p < p' < \infty$. Show that $L_p[0,1] \supset L_{p'}[0,1]$, the inclusion running the other way than in Exercise 2.11.

Solution. Work with the representatives of Remark 2.9, so that $f \in L_{p'}[0,1]$ means $\int_0^1 |f|^{p'} < \infty$. Split the domain according to the size of $|f|$:

$$\int_0^1 |f|^p = \int_{\{|f|>1\}} |f|^p + \int_{\{|f|\le 1\}} |f|^p \le \int_0^1 |f|^{p'} + 1 < \infty.$$

On the first set $|f|^p \le |f|^{p'}$, since $p < p'$ and $|f| > 1$ there; the second integral is at most the measure of $[0,1]$, which is 1. Hence $f \in L_p[0,1]$. ◊

3. Equivalent norms

Definition 3.1 (Equivalent norms)**.** Two norms $\|\cdot\|_a$ and $\|\cdot\|_b$ on a vector space X are *equivalent* if there are constants $c, C > 0$ such that $c\|x\|_a \le \|x\|_b \le C\|x\|_a$ for all $x \in X$.

Equivalent norms induce the same topology, hence the same convergent sequences with the same limits, and the same Cauchy sequences. As a rule of thumb, the norms occurring naturally on an infinite-dimensional space are not equivalent; the finite-dimensional case is the opposite extreme.

Lemma 3.2. Let $(X, \|\cdot\|)$ be a normed space over $\mathbb{K}$ and let $x_1, \dots, x_n \in X$ be linearly independent. Then there is a $C > 0$ such that

$$\Big\|\sum_{i=1}^n \alpha_i x_i\Big\| \;\ge\; C \sum_{i=1}^n |\alpha_i| \qquad \text{for all } \alpha_1, \dots, \alpha_n \in \mathbb{K}.$$

Proof. Both sides are positively homogeneous of degree one in $(\alpha_1, \dots, \alpha_n)$, and both vanish when every α_i is zero, so it suffices to find a $C > 0$ that works for all coefficient vectors with $\sum_i |\alpha_i| = 1$.

Suppose no such C exists. Then for every $m \in \mathbb{N}$ there are coefficients $\alpha_1^{(m)}, \dots, \alpha_n^{(m)}$ with $\sum_i |\alpha_i^{(m)}| = 1$ and

$$\Big\|\sum_{i=1}^n \alpha_i^{(m)} x_i\Big\| < \frac{1}{m}.$$

Each coordinate sequence $(\alpha_i^{(m)})_m$ is bounded by 1, so by the Bolzano–Weierstrass theorem it has a convergent subsequence. Extracting successively for $i = 1, \dots, n$ and passing to the last subsequence, we may assume $\alpha_i^{(m)} \to \alpha_i$ for every i simultaneously. Then $\sum_i |\alpha_i| = \lim_m \sum_i |\alpha_i^{(m)}| = 1$, so not every α_i is zero. The map $(\alpha_1, \dots, \alpha_n) \mapsto \sum_i \alpha_i x_i$ is continuous, and so is the norm, hence

$$\Big\| \sum_{i=1}^{n} \alpha_i x_i \Big\| = \lim_{m \to \infty} \Big\| \sum_{i=1}^{n} \alpha_i^{(m)} x_i \Big\| = 0,$$

so $\sum_i \alpha_i x_i = 0$ with coefficients not all zero. This contradicts the independence of $x_1, \dots, x_n$. ∎

Theorem 3.3. If X is a finite-dimensional vector space over $\mathbb{K}$, then any two norms on X are equivalent.

Proof. Let $x_1, \dots, x_n$ be a basis of X and, for $x = \sum_{i=1}^n \alpha_i x_i$, set $\|x\|_1 = \sum_{i=1}^n |\alpha_i|$, which is a norm on X. Let $\|\cdot\|$ be any norm on X. By Lemma 3.2 there is a $C > 0$ with $\|x\| \geq C\|x\|_1$ for all $x \in X$. In the other direction,

$$\|x\| = \Big\| \sum_{i=1}^{n} \alpha_i x_i \Big\| \leq \sum_{i=1}^{n} |\alpha_i| \|x_i\| \leq \Big(\max_{1 \leq i \leq n} \|x_i\| \Big) \sum_{i=1}^{n} |\alpha_i| = c' \|x\|_1,$$

with $c' = \max_{1 \leq i \leq n} \|x_i\| > 0$, the maximum being positive because the basis vectors are non-zero. So every norm on X is equivalent to $\|\cdot\|_1$.

Equivalence of norms is transitive: if $c_a \|x\|_1 \leq \|x\|_a \leq C_a \|x\|_1$ and $c_b \|x\|_1 \leq \|x\|_b \leq C_b \|x\|_1$ for all x, then

$$\frac{c_b}{C_a} \|x\|_a \leq \|x\|_b \leq \frac{C_b}{c_a} \|x\|_a.$$

Applying this to two arbitrary norms on X, each equivalent to $\|\cdot\|_1$ by the first part, gives the claim. ∎

4. Separable spaces

Definition 4.1 (Countable set)**.** A set S is *countable* if it is finite or can be listed as $S = \{s_1, s_2, s_3, \dots\}$.

Definition 4.2 (Dense set)**.** A set $S \subset (X, d)$ is *dense* if $\overline{S} = X$.

Definition 4.3 (Separable space)**.** A metric space (X, d) is *separable* if it contains a countable dense set.

Remark 4.4. $S \subset (X, d)$ is dense if and only if for every $x \in X$ there is a sequence $(s_n) \subset S$ with $s_n \to x$.

The spaces introduced so far are now checked for completeness and for separability.

Example 4.5. $(\mathbb{R}, |\cdot|)$ is Banach and separable, with $\mathbb{Q}$ as a countable dense set.

Example 4.6. Let $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$. Then $(\mathbb{K}^n, \|\cdot\|_2)$ is Banach and separable: $\mathbb{Q}^n$ is a countable dense set for $\mathbb{K} = \mathbb{R}$, and $(\mathbb{Q} + i\mathbb{Q})^n$ for $\mathbb{K} = \mathbb{C}$.

Example 4.7. $(\mathbb{K}^n, \|\cdot\|_p)$ for $1 \le p < \infty$ and $(\mathbb{K}^n, \|\cdot\|_\infty)$ are Banach and separable, because these norms are equivalent to $\|\cdot\|_2$ by Theorem 3.3, and equivalent norms have the same convergent sequences and the same dense sets.

Remark 4.8. On a finite-dimensional space any such argument may be carried out in the Euclidean norm, since all norms there are equivalent.

Example 4.9. $(\ell_p, \|\cdot\|_p)$ is separable for $1 \le p < \infty$. The countable dense set consists of the finitely supported rational sequences $q = (q_1, \dots, q_m, 0, 0, \dots)$ with $q_i \in \mathbb{Q}$, or with $q_i \in \mathbb{Q} + i\mathbb{Q}$ when $\mathbb{K} = \mathbb{C}$: an $x = (x_1, x_2, \dots) \in \ell_p$ is approximated by truncating it and then approximating the first m coordinates.

Theorem 4.10. $(\ell_p, \|\cdot\|_p)$ is a Banach space for $1 \le p < \infty$.

Proof. Let $(x_m) \subset \ell_p$ be Cauchy, $x_m = (\xi_{1,m}, \xi_{2,m}, \dots)$. Given $\varepsilon > 0$ there is an N such that

$$\Big(\sum_{j=1}^{\infty} |\xi_{j,m} - \xi_{j,n}|^p\Big)^{1/p} \le \varepsilon \qquad \text{for all } n, m > N.$$

A single term is at most the whole sum, so $|\xi_{j,m} - \xi_{j,n}| \le \varepsilon$ for every j and all $n, m > N$. Thus each coordinate sequence $(\xi_{j,m})_m$ is Cauchy in $\mathbb{K}$ and converges; write $\xi_j = \lim_{m\to\infty} \xi_{j,m}$ and $x = (\xi_1, \xi_2, \dots)$.

Fix k. For all $n, m > N$ the partial sum satisfies $\sum_{j=1}^{k} |\xi_{j,m} - \xi_{j,n}|^p \le \varepsilon^p$. Fix $m > N$ and let $n \to \infty$; the sum is finite, so the limit may be taken termwise, and

$$\sum_{j=1}^{k} |\xi_{j,m} - \xi_j|^p \le \varepsilon^p.$$

The bound does not depend on k, so letting $k \to \infty$ gives $\sum_{j=1}^{\infty} |\xi_{j,m} - \xi_j|^p \le \varepsilon^p$, that is $x_m - x \in \ell_p$ and $\|x_m - x\|_p \le \varepsilon$ for every $m > N$.

Consequently $x = (x - x_m) + x_m$ lies in ℓ_p, by Theorem 2.7 and $x_m \in \ell_p$. The estimate $\|x_m - x\|_p \le \varepsilon$ for all $m > N$ is convergence in the norm, not merely coordinatewise, so $x_m \to x$ in ℓ_p. ∎

Counterexample 4.11. $(C[0,1], \|\cdot\|_p)$ is not complete, for $1 \le p < \infty$.

Discussion. A first attempt takes $f_n(x) = x^n$. These are continuous, and

$$\|f_n - 0\|_p = \Big(\int_0^1 x^{np} \mathrm{d}x\Big)^{1/p} = \Big(\frac{1}{np+1}\Big)^{1/p} \longrightarrow 0,$$

so (f_n) is Cauchy but converges in $\|\cdot\|_p$ to the function 0, which is continuous. The attempt establishes nothing: a Cauchy sequence whose limit is continuous is no obstruction to completeness.

Take instead the ramps

$$f_n(x) = \begin{cases} 0, & 0 \le x \le \frac{1}{2} - \frac{1}{n}, \\ n\big(x - \frac{1}{2} + \frac{1}{n}\big), & \frac{1}{2} - \frac{1}{n} \le x \le \frac{1}{2}, \\ 1, & \frac{1}{2} \le x \le 1, \end{cases} \qquad n \ge 3.$$

For $n < m$ the functions f_n and f_m agree outside $[\frac{1}{2} - \frac{1}{n}, \frac{1}{2}]$ and differ by at most 1 on it, so $\|f_n - f_m\|_p \le n^{-1/p}$ and (f_n) is Cauchy in $\|\cdot\|_p$.

Suppose $g \in C[0,1]$ satisfied $\|f_n - g\|_p \to 0$. Fix $\delta > 0$. For $n > 1/\delta$ the function f_n vanishes on $[0, \frac{1}{2} - \delta]$, so

$$\int_0^{1/2-\delta} |g|^p \le \|f_n - g\|_p^p \longrightarrow 0,$$

whence $g \equiv 0$ on $[0, \frac{1}{2} - \delta]$, and since $\delta > 0$ was arbitrary, $g \equiv 0$ on $[0, \frac{1}{2})$; continuity gives $g(\frac{1}{2}) = 0$. The same estimate on $[\frac{1}{2}, 1]$, where $f_n \equiv 1$, gives $g \equiv 1$ there, so $g(\frac{1}{2}) = 1$. This is a contradiction, so no continuous limit exists. ◊

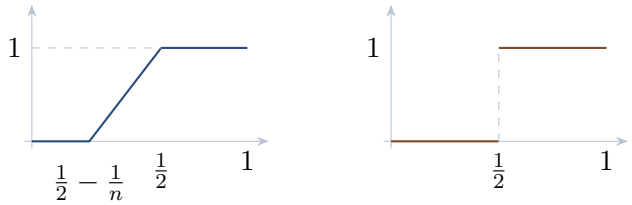


Figure 4.1. *The ramps f_n and the step function they approach. The step function is not continuous, so the limit leaves $C[0,1]$.*

Example 4.12. $L_p[0,1]$ and $L_p(\mathbb{R})$ are Banach and separable for $1 \le p < \infty$. For $L_p[0,1]$ either description may be used: the completion of $(C[0,1], \|\cdot\|_p)$, or the Lebesgue measurable f with $(\int |f|^p)^{1/p} < \infty$. Separability comes from the polynomials with rational coefficients, $p(x) = q_m x^m + \cdots + q_1 x + q_0$ with $q_i \in \mathbb{Q}$, which are dense by Bernstein's theorem.

Theorem 4.13. $(C[0,1], \|\cdot\|_\infty)$ is a Banach space.

Proof. Let $(f_n) \subset C[0,1]$ be a Cauchy sequence. Given $\varepsilon > 0$ there is an N such that $\|f_n - f_m\|_\infty = \max_{x\in[0,1]} |f_n(x) - f_m(x)| < \varepsilon$ for all $n, m > N$, and hence $|f_n(x) - f_m(x)| < \varepsilon$ for every x. So for fixed $x \in [0,1]$ the sequence $(f_n(x))$ is Cauchy in $\mathbb{K}$ and converges; write $f(x) = \lim_{n\to\infty} f_n(x)$. It remains to show that f is continuous and that $f_n \to f$ uniformly.

Let $n > N$. For all $m > N$ one has $|f_n(x) - f_m(x)| < \varepsilon$; letting $m \to \infty$ gives $|f_n(x) - f(x)| \le \varepsilon$ for all $n > N$ and all $x \in [0,1]$, which is uniform convergence.

Now fix $x \in [0,1]$ and $\varepsilon > 0$, and choose N with $|f_N(z) - f(z)| < \frac{\varepsilon}{3}$ for all $z \in [0,1]$. The function f_N is continuous on the compact interval $[0,1]$, hence uniformly continuous, so there is a $\delta > 0$ with $|f_N(x) - f_N(y)| < \frac{\varepsilon}{3}$ whenever $|x - y| < \delta$. For such x, y,

$$|f(x) - f(y)| \le |f(x) - f_N(x)| + |f_N(x) - f_N(y)| + |f_N(y) - f(y)| < \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} = \varepsilon,$$

so $f \in C[0,1]$. ■

Example 4.14. $(C[0,1], \|\cdot\|_\infty)$ is also separable: the separability argument of Example 4.12, with polynomials with rational coefficients, applies here as well.

Example 4.15. Let $C_p^r[0,1]$ carry $\|f\| := \|f\|_p + \|f'\|_p + \cdots + \|f^{(r)}\|_p$. This is a norm, but the space is not complete. Its completion is the Sobolev space $L_p^r[0,1]$.

5. Compact sets

Definition 5.1 (Compactness)**.** A set $K \subset (X,d)$ is *compact* if every open covering of K admits a finite subcovering.

Theorem 5.2. For $K \subset (X,d)$ the following are equivalent.

1. K is compact.
2. Every sequence in K has a convergent subsequence whose limit lies in K.
3. K is complete and totally bounded, that is, for every $\varepsilon > 0$ there are finitely many balls $B(x_1,\varepsilon),\dots,B(x_N,\varepsilon)$ with $K \subset \bigcup_{i=1}^{N} B(x_i,\varepsilon)$.

Lemma 5.3. A compact $K \subset (X,\|\cdot\|)$ is closed and bounded.

Proof. For closedness, let $(x_n) \subset K$ with $x_n \to x$. By Theorem 5.2 the sequence has a subsequence converging to a point of K, and that limit is x, so $x \in K$.

For boundedness, suppose K were unbounded. Then there is a sequence $(x_n) \subset K$ with $\|x_n\| \geq n$, which has no convergent subsequence, contradicting Theorem 5.2. ∎

Theorem 5.4 (Heine–Borel)**.** A set $K \subset \mathbb{R}^n$ is compact if and only if it is closed and bounded.

Definition 5.5 (Local compactness)**.** A metric space (X,d) is *locally compact* if every $x \in X$ has a neighbourhood $B(x,\varepsilon)$ whose closure $\overline{B(x,\varepsilon)}$ is compact.

Definition 5.6 (Relative compactness)**.** A set $S \subset (X,d)$ is *relatively compact* if $\overline{S}$ is compact.

Lemma 5.7 (Riesz)**.** Let $M \subset (X,\|\cdot\|)$ be a proper closed subspace and let $t \in (0,1)$. Then there is an $x_t \in X$ with $\|x_t\| = 1$ and $\operatorname{dist}(x_t, M) \geq t$, where $\operatorname{dist}(x,M) = \inf\{\|x-m\| : m \in M\}$.

Proof. Since M is a proper subspace there is a $u \in X \setminus M$, and since M is closed, $d := \operatorname{dist}(u,M) > 0$. As $t \in (0,1)$ we have $d/t > d$, so by the definition of the infimum there is an $x_0 \in M$ with

$$d \leq \|u - x_0\| \leq \frac{d}{t}.$$

Put $x_t = (u-x_0)/\|u-x_0\|$, a unit vector. Let $x \in M$ be arbitrary and set $x_1 = \|u-x_0\|\,x + x_0$, which lies in M because M is a subspace. Then

$$\|x - x_t\| = \frac{1}{\|u-x_0\|}\big\|\|u-x_0\|\,x - u + x_0\big\| = \frac{\|x_1 - u\|}{\|u-x_0\|} \geq \frac{d}{\|u-x_0\|} \geq \frac{d}{d/t} = t,$$

using $\|x_1 - u\| \geq \operatorname{dist}(u,M) = d$. As $x \in M$ was arbitrary, $\operatorname{dist}(x_t, M) \geq t$. ∎

Theorem 5.8. A normed space $(X, \|\cdot\|)$ is locally compact if and only if $\dim X < \infty$.

Proof. For $x \in X$ and $\varepsilon > 0$ the map $y \mapsto x + \varepsilon y$ is a homeomorphism of X carrying $\overline{B(0,1)}$ onto $\overline{B(x,\varepsilon)}$. All closed balls are therefore compact together with $\overline{B(0,1)}$, and X is locally compact if and only if $\overline{B(0,1)}$ is compact.

Assume $\dim X = \infty$. Choose a unit vector $x_1 \in X$ and put $M_1 = \operatorname{span}\{x_1\}$. Each $M_i = \operatorname{span}\{x_1, \dots, x_i\}$ is finite-dimensional, hence closed: by Theorem 3.3 its norm is equivalent to the norm transported from $\mathbb{K}^i$, which is complete, so M_i is complete and therefore closed in X. It is proper because $\dim X = \infty$. So Lemma 5.7 with $t = \frac{1}{2}$ yields a unit vector x_{i+1} with $\operatorname{dist}(x_{i+1}, M_i) \geq \frac{1}{2}$. This produces a sequence of unit vectors with

$$\|x_n - x_m\| \geq \tfrac{1}{2} \qquad \text{for } n \neq m,$$

since for $n > m$ one has $x_m \in M_{n-1}$. The sequence lies in $\overline{B(0,1)}$ and has no convergent subsequence, so $\overline{B(0,1)}$ is not compact by Theorem 5.2, and X is not locally compact.

Assume $\dim X = n < \infty$. Fixing a basis identifies X with $\mathbb{K}^n$, and by Theorem 3.3 the norm of X is equivalent to the Euclidean norm transported by that identification, so the two have the same closed sets and the same bounded sets. The set $\overline{B(0,1)}$ is closed and bounded, hence compact by Theorem 5.4, and X is locally compact. ∎

Corollary 5.9. $\overline{B(0,1)} \subset (X, \|\cdot\|)$ is compact if and only if $\dim X < \infty$.

6. Compact sets in $(C[a,b], \|\cdot\|_\infty)$

Definition 6.1 (Uniform boundedness)**.** A set $S \subset C[a,b]$ is *uniformly bounded* if there is a $K > 0$ with $\|f\|_\infty \leq K$ for every $f \in S$.

Definition 6.2 (Uniform equicontinuity)**.** A set $S \subset C[a,b]$ is *uniformly equicontinuous* if for every $\varepsilon > 0$ there is a $\delta > 0$ such that for every $f \in S$ and all $x, y \in [a,b]$ with $|x - y| < \delta$ one has $|f(x) - f(y)| < \varepsilon$.

Theorem 6.3 (Arzelà–Ascoli)**.** A set $S \subset (C[a,b], \|\cdot\|_\infty)$ is relatively compact if and only if it is uniformly bounded and uniformly equicontinuous. If S is in addition closed, it is compact.

Proof. Assume first that S is relatively compact. Then $\overline{S}$ is compact, hence bounded by Lemma 5.3, so S is uniformly bounded. Fix $\varepsilon > 0$. By Theorem 5.2 the compact set $\overline{S}$ is totally bounded, so there are finitely many $f_1, \dots, f_r$ with $\overline{S} \subset \bigcup_{i=1}^{r} B(f_i, \frac{\varepsilon}{3})$. Each f_i is continuous on the compact interval $[a,b]$, hence uniformly continuous, so there is a $\delta_i > 0$ with $|f_i(x) - f_i(y)| < \frac{\varepsilon}{3}$ whenever $|x - y| < \delta_i$. Put $\delta = \min_{1 \leq i \leq r} \delta_i$, which is positive because the minimum is over finitely many indices. Given $f \in S$, choose i with $\|f - f_i\|_\infty < \frac{\varepsilon}{3}$. Then for $|x - y| < \delta$,

$$|f(x) - f(y)| \leq |f(x) - f_i(x)| + |f_i(x) - f_i(y)| + |f_i(y) - f(y)| < \varepsilon,$$

so S is uniformly equicontinuous.

Assume now that S is uniformly bounded, by K, and uniformly equicontinuous, and let $(f_n) \subset S$. The goal is to select a convergent subsequence. Enumerate the rationals of the interval,

$\mathbb{Q} \cap [a,b] = \{x_1, x_2, \ldots\}$. The values $f_n(x_1)$ satisfy $|f_n(x_1)| \le K$, so some subsequence $(f_n^{(1)})$ converges at x_1. Along it, the values at x_2 are again bounded, so a further subsequence $(f_n^{(2)})$ converges at x_2, and so on. Let $g_m = f_m^{(m)}$ be the diagonal sequence. It is a subsequence of the original (f_n), and from the index k onwards it is a subsequence of $(f_n^{(k)})$, so $(g_m(x_k))$ converges for every k.

The claim is that (g_m) is uniformly Cauchy. Fix $\varepsilon > 0$ and let $\delta > 0$ be given by uniform equicontinuity for $\frac{\varepsilon}{3}$. Cover $[a,b]$ by finitely many sets $U_1, \ldots, U_s$ of diameter less than δ, and pick in each U_j a rational point x_{k_j}, which exists because the rationals are dense. For each j the sequence $(g_m(x_{k_j}))_m$ converges, hence is Cauchy, so there is an N_j with $|g_n(x_{k_j}) - g_m(x_{k_j})| < \frac{\varepsilon}{3}$ for $n, m > N_j$. Put $N = \max_{1 \le j \le s} N_j$, finite because there are finitely many j. Let $t \in [a,b]$ and choose j with $t \in U_j$; then $|t - x_{k_j}| < \delta$, so for $n, m > N$,

$$|g_n(t) - g_m(t)| \le |g_n(t) - g_n(x_{k_j})| + |g_n(x_{k_j}) - g_m(x_{k_j})| + |g_m(x_{k_j}) - g_m(t)| < \varepsilon.$$

Hence $\|g_n - g_m\|_\infty \le \varepsilon$ for $n, m > N$. The space $(C[a,b], \|\cdot\|_\infty)$ is complete, by the proof of Theorem 4.13 with $[a,b]$ in place of $[0,1]$, so (g_m) converges in it.

Finally, $\overline{S}$ has the same uniform bound K and the same modulus of equicontinuity as S, so the selection above applies verbatim to a sequence in $\overline{S}$, and its limit lies in $\overline{S}$ because $\overline{S}$ is closed. By Theorem 5.2 the set $\overline{S}$ is compact, that is, S is relatively compact. ∎